Name: ____________________

ANGLES AND GEOMETRY, (ANGLES IN PARALLEL LINES)

Date: ________________

LO: I can identify and calculate alternate, corresponding and co-interior angles in parallel lines.

Angle relationships in parallel lines

When a **transversal** crosses two **parallel lines**, special angle pairs are formed: alternate, corresponding, and co-interior angles.

Remember: Alternate angles are equal (Z-shape). Corresponding angles are equal (F-shape). Co-interior angles add up to 180°.

Angle facts

●	Alternate angles are equal	Z-angles
●	Corresponding angles are equal	F-angles
●	Co-interior angles sum to 180°	C/U-angles
●	Co-interior angles are equal	they sum to 180°
●	Alternate angles add to 180°	they are equal

Key idea

Look for the Z, F, or C/U shape to identify the angle relationship. Then use equal or sum to 180°.

Alternate = equal; Co-interior = 180°

Z = equal, F = equal, C = 180°

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – alternate angles

Two parallel lines are cut by a transversal. One angle is 65°. Find the alternate angle.

Identify the Z-shape

Alternate angles are equal

The angle = 65°

= 65°

Alternate angles are always equal when lines are parallel.

Example 2 – corresponding angles

A transversal crosses parallel lines. One angle is 110°. Find the corresponding angle.

Identify the F-shape

Corresponding angles are equal

The angle = 110°

= 110°

Corresponding angles are in the same position at each intersection.

Example 3 – co-interior angles

Two parallel lines are cut by a transversal. One co-interior angle is 72°. Find the other.

Co-interior angles sum to 180°

180 - 72 = 108

The other angle = 108°

= 108°

Co-interior (allied) angles always add up to 180°.

Example 4 – finding unknown

In parallel lines, angle x and 3x are co-interior. Find x.

Co-interior angles sum to 180°

$x + 3x = 180 \quad 4x = 180$

$x = 45°$

x = 45°

Form an equation using the angle fact, then solve.

YOUR TURN – PRACTICE (deliberate practice)

1. Alternate angles (Z)

Find the marked angle. Lines are parallel.

a) Alternate to 55°

b) Alternate to 120°

c) Alternate to 83°

d) Alternate to 145°

2. Corresponding angles (F)

Find the corresponding angle.

a) Corresponding to 70°

b) Corresponding to 95°

c) Corresponding to 132°

d) Corresponding to 48°

3. Co-interior angles (C/U)

Find the co-interior angle. They sum to 180°.

a) Co-interior with 60°

b) Co-interior with 115°

c) Co-interior with 78°

d) Co-interior with 143°

4. Finding unknowns

Find the value of x.

a) x and 2x are alternate angles

b) x and (180 - x) are co-interior

c) 2x + 10 and 70 are corresponding

d) 3x and 2x + 40 are alternate

5. Mixed angle problems

State the relationship and find the missing angle.

a) Angle a is alternate to 74°

b) Angle b is co-interior with 105°

c) Angle c is corresponding to 138°

d) x and 4x - 20 are co-interior; find x

e) 2x + 15 and 3x - 10 are alternate; find x

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Alternate angles are equal ☐

Correct answer: __________

Reason: _________________

b) Co-interior angles are equal ☐

Correct answer: __________

Reason: _________________

c) Corresponding angles are equal ☐

Correct answer: __________

Reason: _________________

d) Alternate angles add to 180° ☐

Correct answer: __________

Reason: _________________

e) These rules only work with parallel lines ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Two parallel lines are cut by a transversal. One angle is **(3x + 15)°** and its alternate angle is **(5x - 25)°**. Find x and both angles.

Expression: ____________________________

Simplified: ____________________________

b) A transversal crosses two parallel lines creating 8 angles. If one angle is **62°**, find all 8 angles. Explain your reasoning using the correct angle vocabulary.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!